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Q-HAS: Quantum-Hierarchical Adaptive Steering

A Predictive Hybrid Framework for Topological
Instability Detection in Plasma

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Agenda

- 1 Motivation
- 2 Abstract
- 3 Our approach
- 4 What needs to be adapted
- 5 Theoretical results
- 6 Computational results
- 7 Discussion and Conclusions

1. Motivation

Nuclear

1 Renewable Energy with high yield production

2 Nuclear Fusion could be a solution

3 Perform a Simulation of Nuclear Fusion

4 Classical AMR method is too reactive

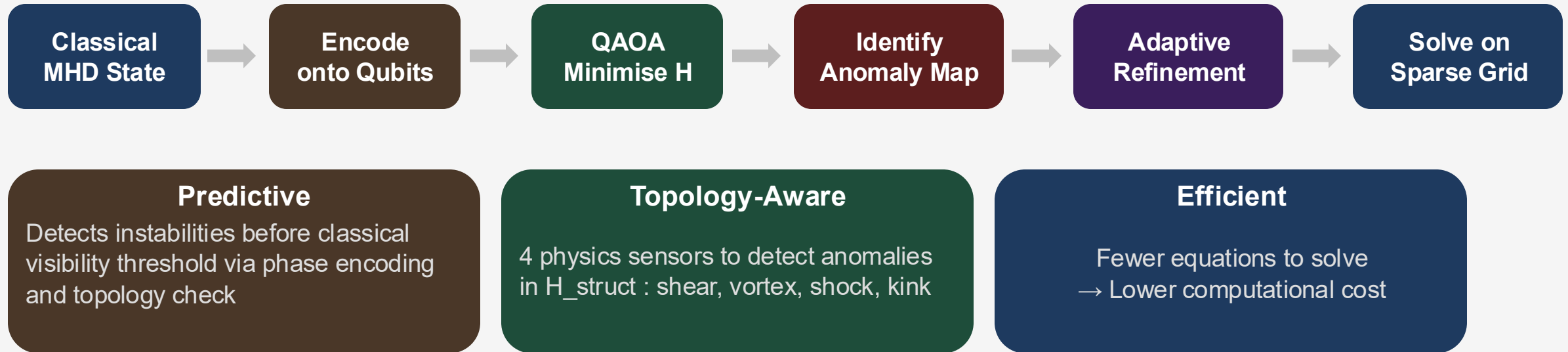


4_{bis}

Q-HAS : Quantum AMR **predictive**

2. What is Q-HAS ?

Hybrid Quantum–Classical Framework for Predictive AMR



Details about Q-HAS

Qubits – Operators – Algorithm

Q-HAS encodes a physical state into a quantum state to predict the behaviour of our system.

3. What makes it different from the classic AMR ?

3.1 Qubits and Quantum Phase information

Dual Flux Graph:

Domain $\rightarrow$ coarse blocks $\rightarrow$ graph $G = (V, E)$

Each edge of E = stress flux ∇U between cells = One qubit

Bloch Sphere Encoding:

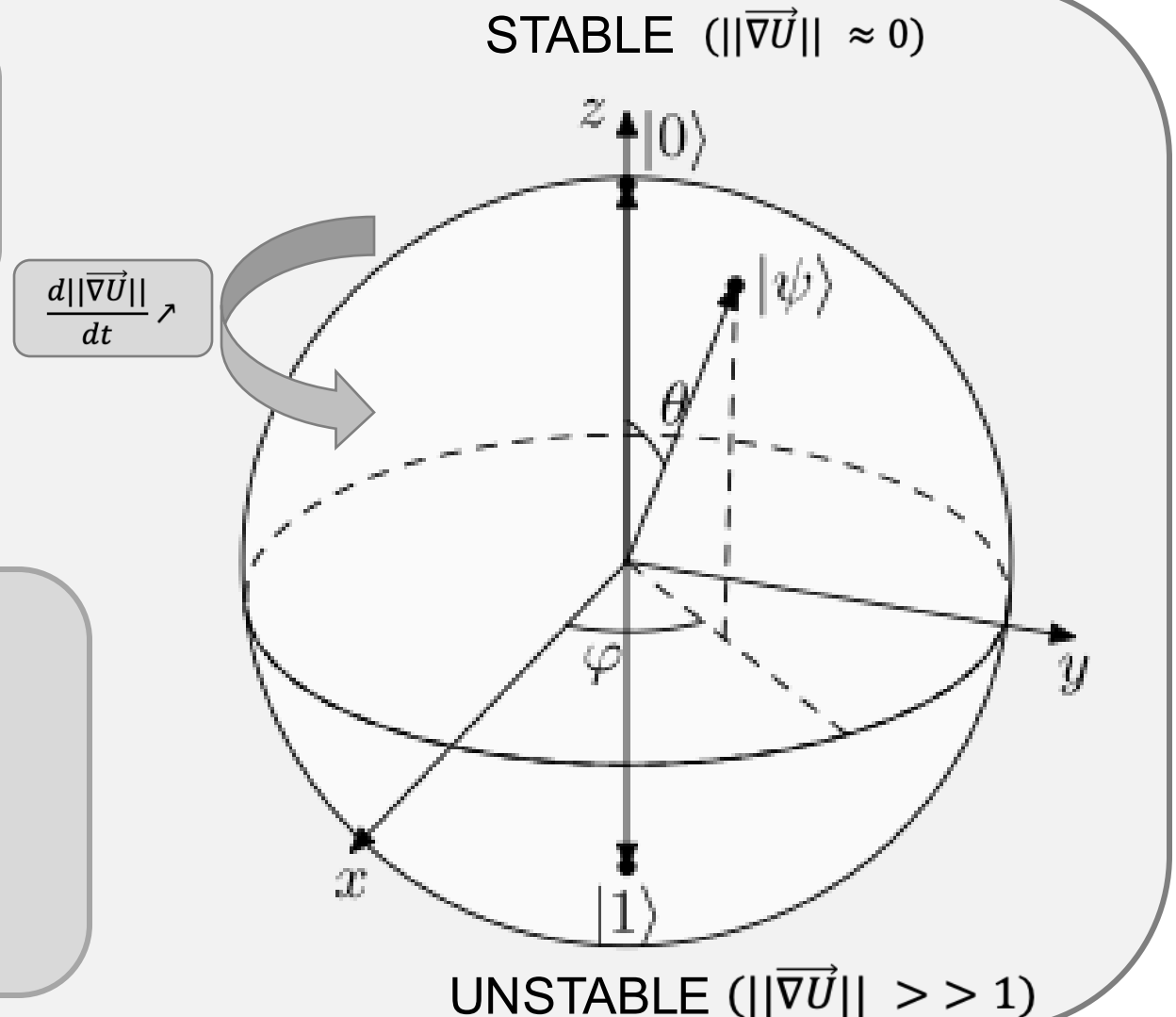
$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\varphi} \sin\left(\frac{\theta}{2}\right) |1\rangle$$

θ (polar) = flux amplitude $\theta \propto ||\nabla U||$

High $\theta \rightarrow$ qubit near $|1\rangle$ (unstable)

φ (azimuthal) = growth rate $\varphi \propto \frac{d||\nabla U||}{dt}$

High $\varphi \rightarrow$ rotation around z-axis (rate of instability)



3. What makes it different from the classic AMR ?

3.2 The Hamiltonian of Instability : H_{struct}

$$\text{In 2D : } H_{struct} = \sum_i h_i Z_i + \sum_{\langle i,j \rangle} C_{ij} Z_i Z_j + \sum_P K_P \prod_{l \in P} Z_l + \sum_V \Delta_V \prod_{k \in V} Z_k$$

Z

Activity Bias
Flags high stress flux

ZZ

Gradient Coupling
Detects shear layers

ZZZZ

Circulation
Vortices + current
sheets

ZZZZ

Shock Vertex
Supersonic shocks

XY-YX

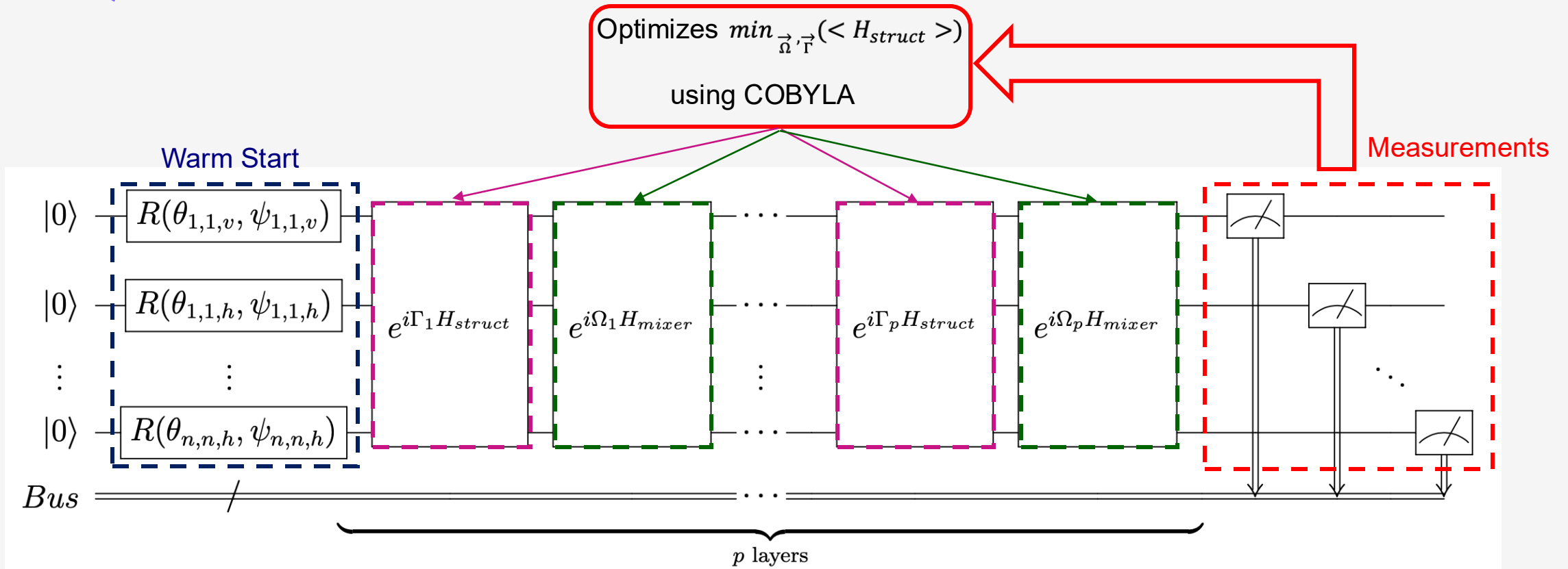
Kink (3D)
Helical twisting

Each term acts as a specific physics sensor → Ground state of H_{struct} identifies anomaly pattern matching known MHD instabilities

For quantum tunneling and quantum fluctuations, we also need : $H_{mixer} = \prod_i X_i$

3. What makes it different from the classic AMR ?

3.3 The QAOA Variational Ansatz



Warm Start: qubits initialised from physical data **avoids barren plateau**

QAOA alternates p layers of structural Hamiltonian (phase imprint) + Mixer Hamiltonian (tunnelling)

$\boxed{}$ = Phase Separator (imprints physics) $\boxed{}$ = Mixer (tunnelling + interference)

4. What needs to be implemented to practice it

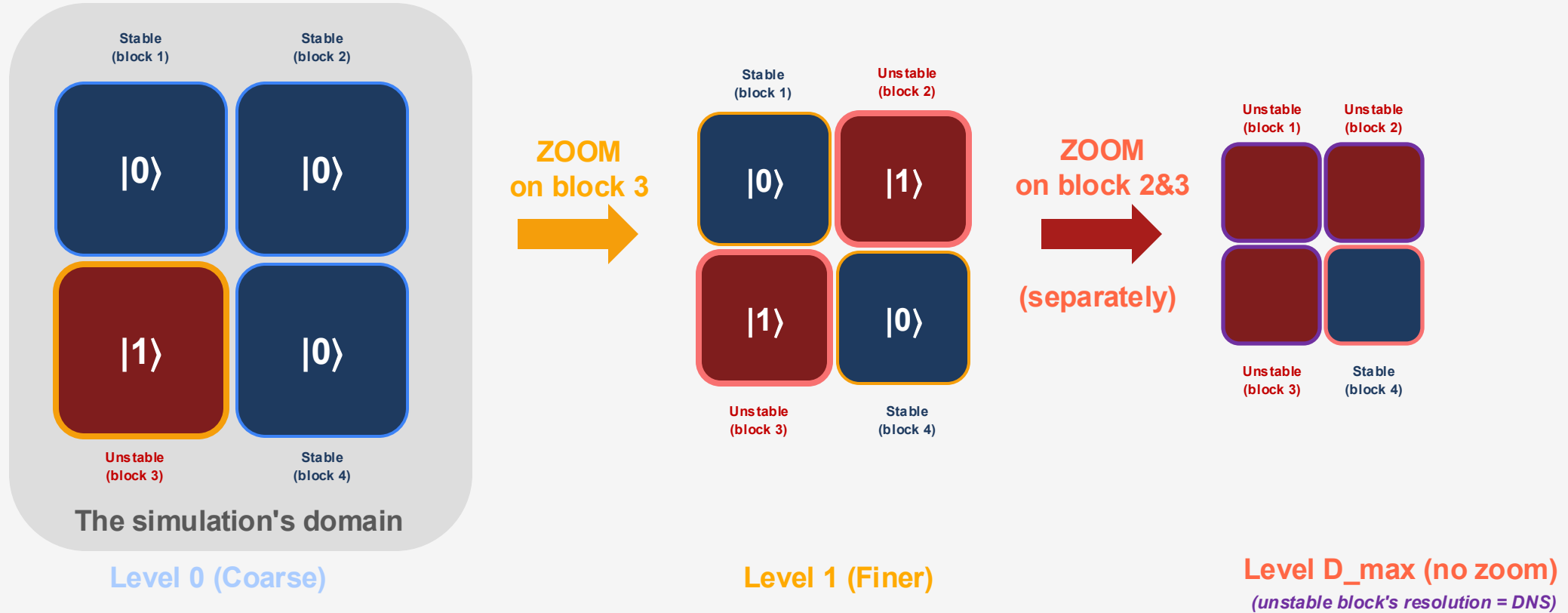
Limitations

1. Number of qubits
2. Depth of our quantum circuit.
3. Resolution techniques we can use to solve plasma equations when the mesh is adaptative

4. What needs to be implemented to practice it

4.1 Limitation on the number of qubits

Fixed qubit budget $\rightarrow$ hierarchical BFS: coarse scan $\rightarrow$ zoom into unstable regions $\rightarrow$ recurse



4. What needs to be implemented to practice it

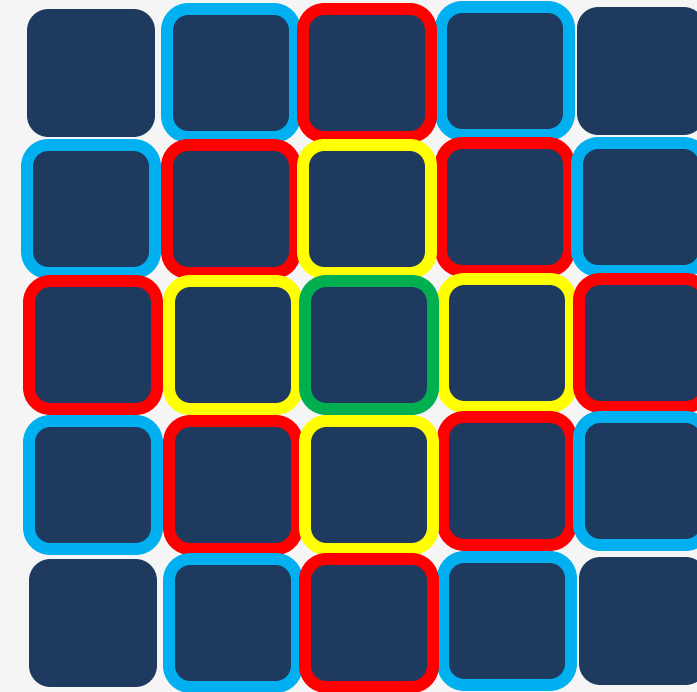
4.2 Circuit Depth and Receptive field (Balancing Spatial Awareness vs. Decoherence)

If we analyze the vision of qubit 
based on the number of layers of the QAOA:

 $p = 1 \rightarrow$ Local vision (like classical stencil)
Only immediate neighbours

 $p = 3$  $p = 2 \rightarrow$ Non-local awareness
Detects filaments, X-points, vortices

Optimal: $p \approx$ graph diameter $d_{\max}$
Beyond $d_{\max}$: no new info, only noise
generated by a bigger circuit's depth



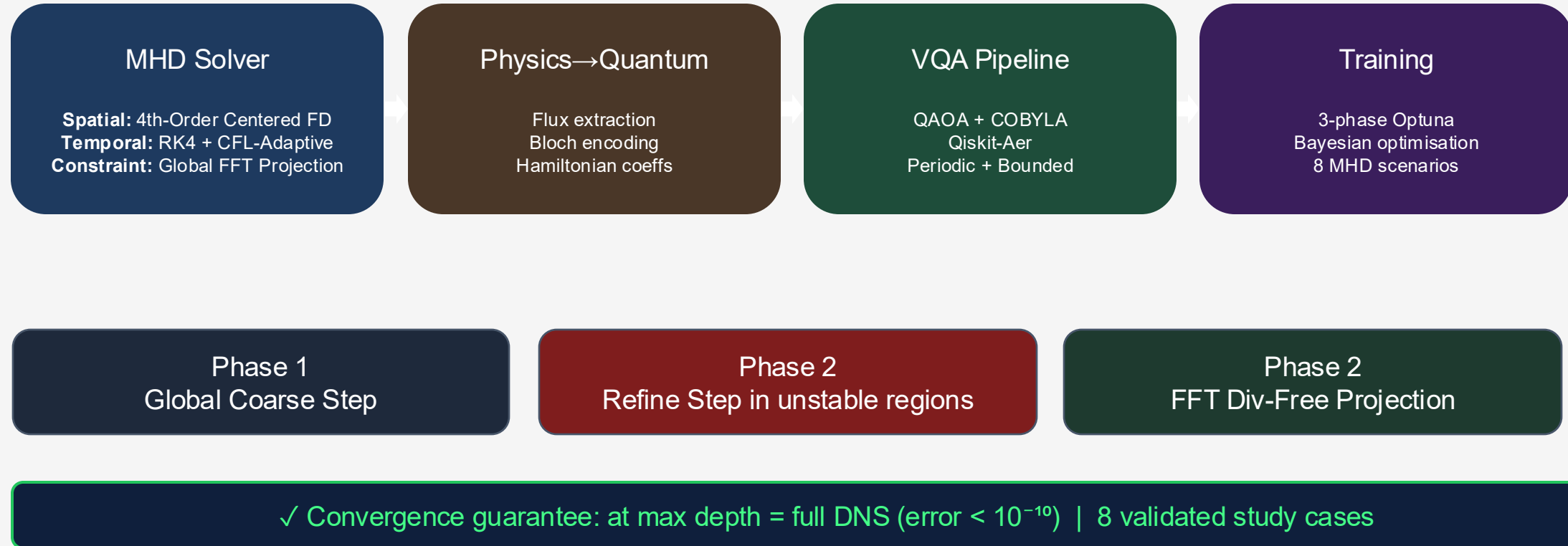
"Receptive field" depending on the
number of layer p of the QAOA

*To avoid noise and limit the depth of the circuit we also perform:
Gate pruning (remove gates below threshold)*

4. What needs to be implemented to practice it

4.3 Numerical Methods and Software Stack (FD4 + RK4 + FFT Projection)

Complete hybrid software stack: Classical MHD + Quantum VQA + Training pipeline



Results of Q-HAS

Theoretical and computational results

5. Theoretical Results

5.1 What is happening with the QAOA in this minimization?

Two key proofs: (1) optimisation cannot start without mixer (2) force has dual structure

At initialization ($\vec{\Omega} = 0$):

Null-Gradient Proposition

$$\nabla_{\vec{\Gamma}} E = i \partial_{\Gamma_k} \langle \psi_0 | H_{struct} | \psi_0 \rangle = 0 \quad \forall k$$

Algorithm MUST activate the mixer (Ω) to break symmetry and start optimising

Mixer Gradient (Driving Force)

$$\nabla_{\vec{\Omega}} E = i \partial_{\Omega_k} \langle \psi_k | [H_{struct}, H_{mixer}] | \psi_k \rangle$$

Phase imprint from H_{struct} creates non-zero gradient $\rightarrow$ instant ignition

$$F_q \propto \sin(\theta_q) \sin(\varphi_q) \prod_{m \in P, m \neq q} \cos(\theta_m)$$

Each term has a Phase Boost & Topological coherence

$$\prod_{m \in P, m \neq q} \cos(\theta_m) \approx \begin{cases} 0 & (\text{Noise} \rightarrow \text{no flip}) \\ 1 & (\text{coherent neighbours' structure} \rightarrow \text{flip}) \end{cases}$$

After initialization ($\vec{\Omega} \neq 0$):

$$\text{We need to consider : } H_{eff}^{(k)} = U_{>k}^\dagger H_{struct} U_{>k}$$

Other advantages due to the use of QAOA :

Structural Gradient (ensures minima search)

$$F_{\Gamma_k} = -i \langle \psi_k | [H_{struct}, H_{eff}^{(k)}] | \psi_k \rangle$$

Constructive interferences for low energy states
(Rotation around Z)

Mixer Gradient (Driving Force)

$$F_{\Omega_k} = -i \langle \psi_k | [H_{mixer}, H_{eff}^{(k)}] | \psi_k \rangle$$

Quantum tunneling to explore the configuration space
(Rotation around X)

5. Theoretical Results

5.2 Complexity of the Q-HAS : Q-HAS vs. DNS vs. Classical AMR

DNS

$$\mathcal{O}(N_{fine}^d)$$

Solve everywhere at full resolution
Cost: $Re^{(9/4)}$ in 3D
→ Intractable at fusion scales

Classical AMR

$$\mathcal{O}(S_{classic} N_{fine}^d T_{physics})$$

Refine where gradients are large
But: reactive → over-refinement

Q-HAS

$$\mathcal{O}(S_{quantum} N_{fine}^d (T_{physics} + \nu_{AMR} T_{VQA}))$$

Predictive topological filtering
Phase detection → fewer scans

← Computational Cost →

DNS:



AMR:



Q-HAS:



Advantage of Q-HAS if :

$$\frac{S_{quantum}}{S_{classic}} < \frac{T_{physics}}{T_{physics} + (T_{physics} + \nu_{AMR} T_{VQA}))}$$

6. Computational Results

6.1 Training results: Three-Phase Bayesian Hyperparameter Optimisation

Phase 1 — Encoding + AMR

$\alpha, \beta, \text{thr_amr}$
Re 400 | 200 trials

J = 0.205
E_{phys} = 11.2%
S = 51.6%

Phase 2 — Hamiltonian Weights

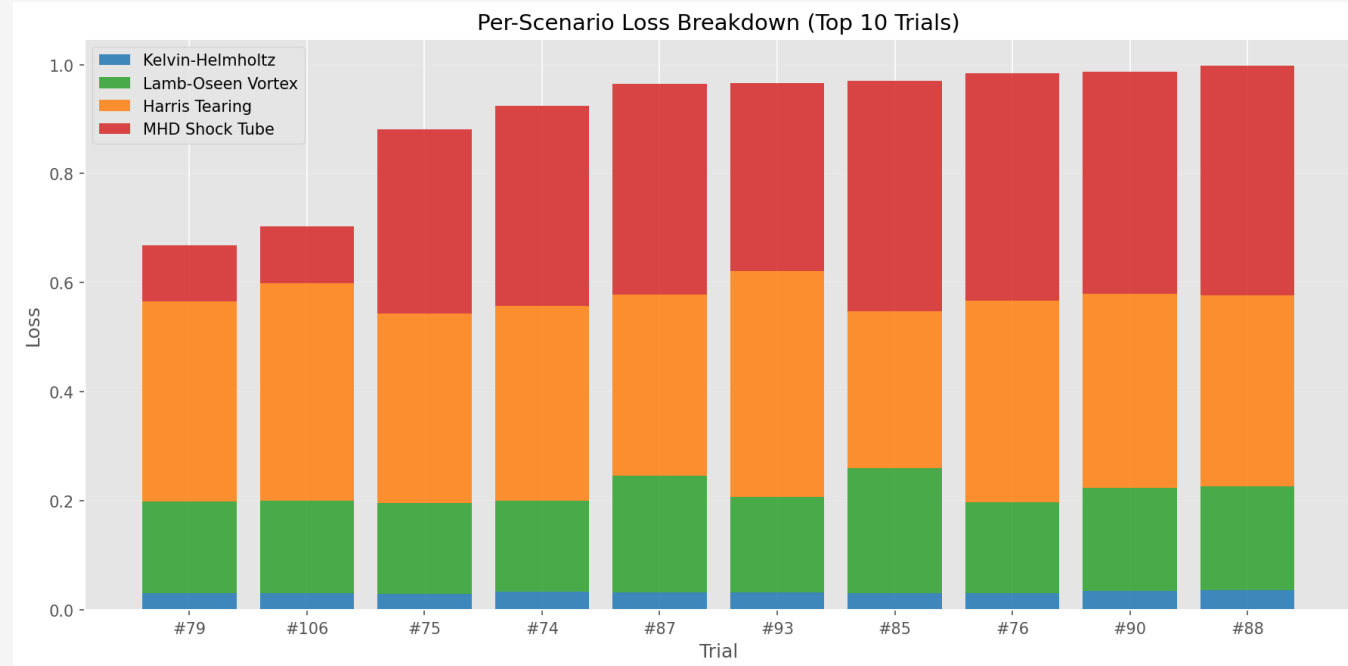
$w_{\text{data}}, w_{\text{shear}}, w_{\text{vortex}}, \gamma_2$
Re 400 | 400 trials

J = 0.220
E_{phys} = 19.7%
S = 29.5%

Phase 3 — Advanced Anomalies

w_{shock} + retrain all
Re 800 | 600 trials

Shock detection
High Re regime
[pending]



6. Computational Results

6.2 Benchmark Comparisons vs. Classical AMR

Three "trap scenarios" designed to expose classical AMR weaknesses

A Phase Sensitivity

Kelvin-Helmholtz

Classical AMR:
Misses linear growth phase
(constant gradient $\rightarrow$ no refine)

Q-HAS:
Detects growing $\partial\Phi/\partial t$
 $\rightarrow$ Pre-emptive refinement

$TTD_QHAS < TTD_classical$

B Noise Immunity

Noisy Uniform Field

Classical AMR:
 ∇^2 amplifies noise
 $\rightarrow$ Flags entire domain ($S \rightarrow 1$)

Q-HAS:
Plaquette $\oint \text{noise} \cdot dl \rightarrow 0$
 $\rightarrow$ Topological low-pass filter

$\eta_{\text{eff}} \approx 1.0$ (0% refine)

C Topological Selectivity

Harris Current Sheet

Classical AMR:
Refines entire sheet
(strong gradients everywhere)

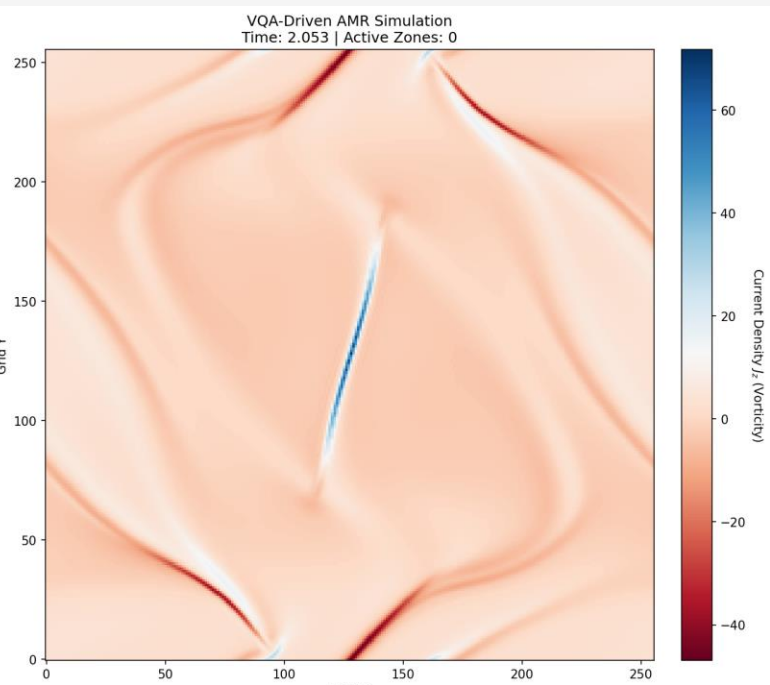
Q-HAS:
Detects only X-points
(topology breaking)

$R_{\text{topo}} = \text{X-pt area} / \text{total}$

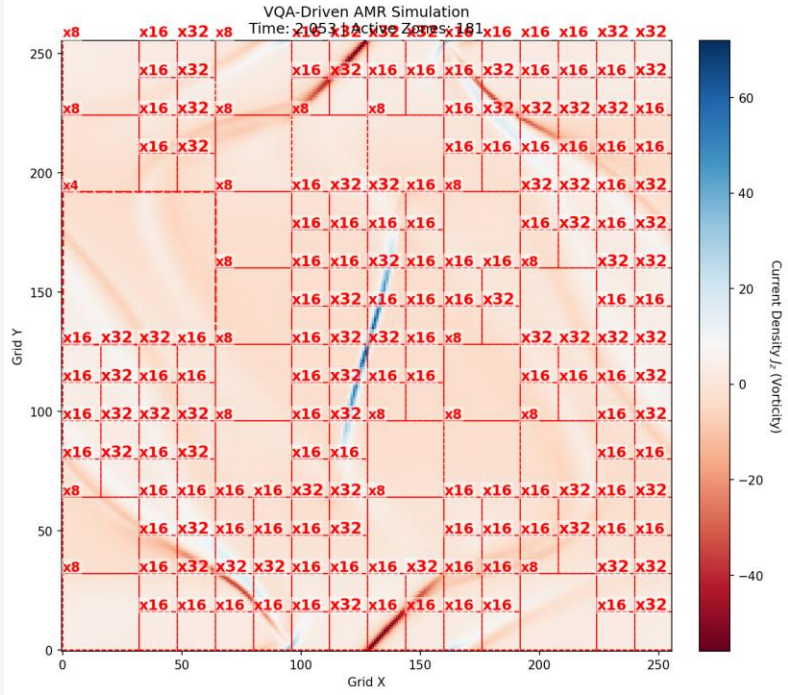
[Benchmark result graphs to be inserted — simulations pending]

6. Computational Results

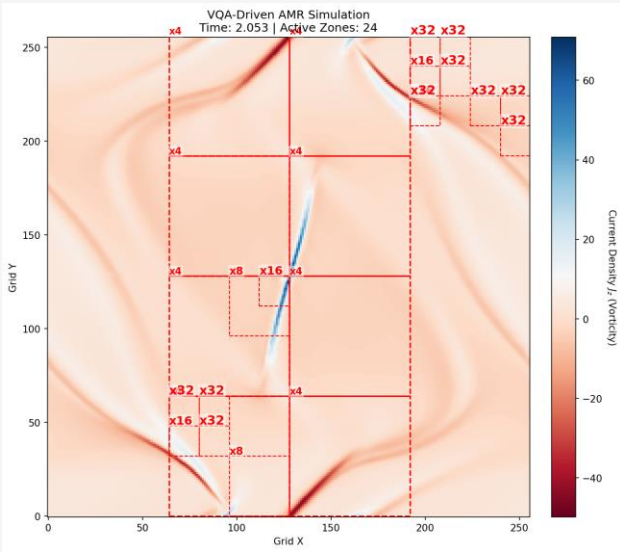
6.3 The v1



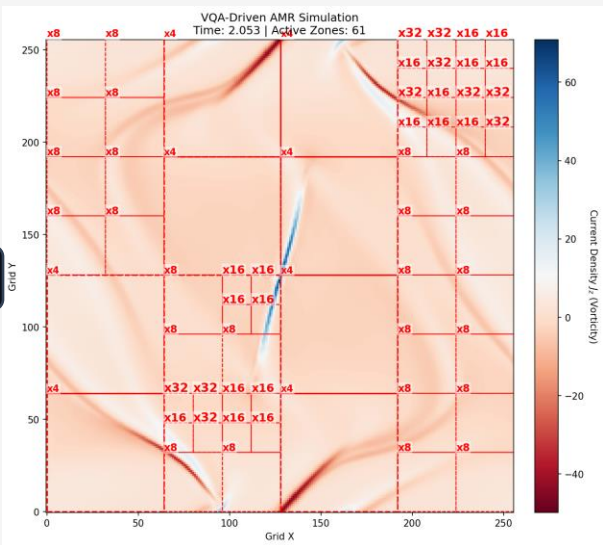
DNS



Classic AMR



Q-HAS



Discussion and conclusions about Q-HAS

7. Discussion & Conclusions

Key Insights, Limitations, and Future Work

Q-HAS: topology-aware quantum steering for classical plasma simulations

Key Theoretical Result

- Phase + topology deeply coupled
- Force = Phase Boost $\times$ Coherence
- Noise: destructive interference $\rightarrow$ filtered

Key Computational Result

- All 8 VQA study cases validated
- Cross-anomaly isolation confirmed
- AMR $\rightarrow$ DNS at max depth ($< 10^{-10}$)

Limitations

- Simulator only (no real quantum HW)
- FD4 non-conservative at boundaries
- Signal saturation in high-energy regime

Future Work

- Real IBM quantum hardware
- Finite Volume Method (conservation)
- 3D extension + kink (DMI) term
- VQA parallelisation across patches

Quantum computing as a specialised complement, not a wholesale replacement

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Thank you